PRACTICAL LAB: JAVASCRIPT OPERATORS MASTERY

Course: DFK20013 Web Design Technology

Topic: 3.1.1 Variables, Data Types, and Operators

Duration: 60 – 90 Minutes

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1. Learning Outcomes

By the end of this lab, students should be able to:

- Implement **Arithmetic Operators** for calculations.
- Use **Assignment Operators** to update values efficiently.
- Control application states using **Increment/Decrement**.
- Validate user access using **Relational** and **Logical** operators.

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2. Practical Task: The All-in-One Operator Workshop

Create a file named lab_operators.html. Copy and complete the coding below.

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <title>DFK20013: JS Operators Workshop</title>

    <style>

        body { font-family: 'Segoe UI', sans-serif; max-width: 850px;
margin: 2rem auto; background-color: #f0f2f5; padding: 20px; color: #333;
}

        section { background: white; padding: 25px; border-radius: 12px;
margin-bottom: 25px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }

        h2 { color: #2c3e50; border-left: 5px solid #3498db; padding-
left: 10px; margin-top: 0; }

        .output-box { background: #2c3e50; color: #27ae60; padding: 15px;
border-radius: 6px; font-family: 'Courier New', monospace; margin-top:
10px; }

        .short-note { background: #fff3cd; color: #856404; padding: 15px;
border-radius: 6px; margin-top: 15px; border-left: 5px solid #ffc107;
display: none; font-size: 0.9rem; }
```

```
        input { padding: 8px; width: 100px; border-radius: 4px; border:
1px solid #ccc; }

        button { padding: 8px 20px; background: #3498db; color: white;
border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }

        button:hover { background: #2980b9; }

        code { background: #eee; padding: 2px 4px; border-radius: 3px;
font-weight: bold; color: #d63384; }

    </style>

</head>
```

```
<body>



    <header style="text-align: center; margin-bottom: 30px;">

        <h1>JavaScript Operators Mastery</h1>

        <p>Follow the steps and observe the Console and UI outputs.</p>

    </header>
```

```
    <section>

        <h2>Task 1: Basic Arithmetic (Check Console)</h2>

        <p>This task demonstrates background calculations using
<code>+</code>, <code>-</code>, <code>*</code>, <code>/</code>,
<code>**</code>, and <code>%</code>.</p>

        <button onclick="runTask1()">Run Console Logic</button>

    </section>
```

```
    <section>

        <h2>Task 2: POS System (Arithmetic & Assignment)</h2>

        <p>Price: RM 50.00. <strong>Promo:</strong> Buy > 4 units to get
RM 20.00 discount.</p>

        Quantity: <input type="number" id="qty" value="1">

        <button onclick="runTask2()">Calculate Total</button>

        <div id="outPOS" class="output-box">Final Total: RM 0.00</div>

        <div id="notePOS" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 3: Stock Control (Increment & Decrement)</h2>

        <p>Simulating inventory updates.</p>
```

```
        Stock Level: <span id="stockVal" style="font-size: 1.5rem; font-
weight: bold;">10</span><br><br>

        <button onclick="runTask3('inc')">Add (+)</button>

        <button onclick="runTask3('dec')"
style="background:#6c757d;">Remove (-)</button>

        <div id="noteStock" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 4: Gate Access (Relational & Logical)</h2>

        <p>Rules: Age 18+ OR (Age 13+ AND Parent present).</p>

        Age: <input type="number" id="age" placeholder="Age">

        <label><input type="checkbox" id="parent"> With
Parent?</label><br><br>

        <button onclick="runTask4()" style="background:#27ae60;">Verify
Entry</button>

        <div id="outSecurity" class="output-box">Status: Waiting...</div>

        <div id="noteSecurity" class="short-note"></div>

    </section>
```

Inside the <script> tag, type all the following code:

JAVASCRIPT:

1.

```
<script>
    // Helper to show explanations
    function showNote(id, text) {
        const noteDiv = document.getElementById(id);
        noteDiv.style.display = 'block';
        noteDiv.innerHTML = text;
    }
```

2.

```
// --- TASK 1: ARITHMETIC (BASIC) ---
function runTask1() {
    console.log("--- Task 1: Arithmetic Operations ---");
    let students = 31;
    students = students + 1;   // Addition
    students = students - 2;   // Subtraction
    students = students * 2;   // Multiplication
    students = students / 2;   // Division
    let power = 10 ** 2;       // Exponentiation (100)
    let remainder = 31 % 2;    // Modulus (1 - Odd)

    console.log("Final Student Count: " + students);
    console.log("10 to the power of 2: " + power);
    console.log("Remainder of 31/2: " + remainder);
    alert("Check Browser Console (F12) for results!");
}
```

3.

```
// --- TASK 2: POS SYSTEM (ARITHMETIC & ASSIGNMENT) ---
function runTask2() {
    let price = 50;
    let qty = parseInt(document.getElementById('qty').value);

    // a. Arithmetic Operator (*)
    let total = price * qty;
    let note = "<b>Subtopic (a) Arithmetic:</b> <code>*</
    code> multiplied " + price + " by " + qty + ".<br>";

    // b. Assignment Operator (-=)
    if (qty > 4) {
        total -= 20; // Shorthand for total = total - 20
        note += "<b>Subtopic (b) Assignment:</b> <code>-=</
        code> updated the total by subtracting the discount.";
    }

    document.getElementById('outPOS').textContent = "Final
    Total: RM " + total.toFixed(2);
    showNote('notePOS', note);
}
```

4.

```
// --- TASK 3: INCREMENT & DECREMENT ---
let stock = 10;
function runTask3(type) {
    let note = "";
    if (type === 'inc') {
        stock++; // (c) Increment
        note = "<b>Subtopic (c) Increment:</b> <code>++</
        code> increased the stock by 1.";
    } else {
        stock--; // (c) Decrement
        note = "<b>Subtopic (c) Decrement:</b> <code>--</
        code> decreased the stock by 1.";
    }
    document.getElementById('stockVal').textContent = stock;
    showNote('noteStock', note);
}
```

5.

```
    // --- TASK 4: RELATIONAL & LOGICAL ---
    function runTask4() {
        let age = parseInt(document.getElementById('age').value);
        let hasParent = document.getElementById('parent').checked;
        let status = document.getElementById('outSecurity');

        if (isNaN(age)) return alert("Please enter a valid age.");

        // d. Relational: >=
        // e. Logical: || (OR), && (AND)
        let isEligible = (age >= 18) || (age >= 13 && hasParent);

        status.textContent = isEligible ? "Status: ACCESS GRANTED
        ✅" : "Status: ACCESS DENIED ❌";
        status.style.color = isEligible ? "#27ae60" : "#ff4d4d";

        let note = "<b>Subtopic (d) Relational:</b> <code>>=</
        code> compared Age (" + age + ") to the limit.<br>";
        note += "<b>Subtopic (e) Logical:</b> Used <code>||</
        code> (OR) and <code>&&</code> (AND) to combine rules.";
        showNote('noteSecurity', note);
    }
</script>
```

Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

1.	**Screenshot of the output** Task 1: Basic Arithmetic (Check Console) This task demonstrates background calculations using +, -, *, /, **, and %. Run Console Logic
	Explanation The function showNote(id, text) is used to display a note or explanation on a webpage. The id parameter represents the ID of the HTML element where the note will appear, while the text parameter contains the message that will be shown. The function first finds the HTML element using document.getElementById(id). Then, it makes the element visible by setting the display style to "block". Finally, it inserts the explanation text into the element using innerHTML. This allows a hidden note or explanation to appear on the webpage when the function is called.

2.	**Screenshot of the output** Task 1: Basic Arithmetic (Check Console) This task demonstrates background calculations using +, -, *, /, **, and %. Run Console Logic
	ExplanationThe function `runTask1()` shows how **basic arithmetic operations** work in JavaScript. First, it prints a message in the console to indicate that Task 1 has started. A variable called `students` is created with the value **31**. The program then performs several calculations such as **addition (+), subtraction (-), multiplication (*), and division (/)** using the `students` variable. Next, the code calculates **10 to the power of 2** and stores the result in the variable `power`. It also finds the **remainder when 31 is divided by 2** using the modulus operator `%`, which is stored in the variable `remainder`. After that, the results are printed in the browser console using `console.log()`. Finally, an alert message appears telling the user to **open the browser console (press F12)** to see the results.

3.	**Screenshot of the output** Task 2: POS System (Arithmetic & Assignment) Price: RM 50.00. **Promo:** Buy > 4 units to get RM 20.00 discount. Quantity: 1 Calculate Total Final Total: RM 0.00
	Explanation The function `runTask2()` is used to simulate a **simple Point of Sale (POS) system**. First, the code sets the **price** of an item to **RM50**. Then, it gets the **quantity (qty)** entered by the user from an input field in the webpage using `document.getElementById('qty').value`. The `parseInt()` function converts the input value into a number. Next, the program calculates the **total price** by multiplying the price with the quantity using the arithmetic operator `*`. A note is also created to explain the multiplication process. After that, the code checks if the quantity is **greater than 4**. If this condition is true, the program applies a **discount of RM20** using the assignment operator `-=`. This operator subtracts 20 from the total amount. Finally, the program displays the **final total price in Ringgit Malaysia (RM)** on the webpage and shows the explanation note using the `showNote()` function.

4.	**Screenshot of the output** Task 4: Gate Access (Relational & Logical) Rules: Age 18+ OR (Age 13+ AND Parent present). Age: Age ☐ With Parent? Verify Entry Status: Waiting...

	Explanation The variable **stock** starts with the value **10**. The function **runTask3(type)** changes the stock value. If **type = "inc"**, the stock **increases by 1** using **stock++**. If the type is not **"inc"**, the stock **decreases by 1** using **stock--**. The new stock value is shown on the webpage. A message is displayed to show whether the stock was increased or decreased.

5.	**Screenshot of the output** Status: ACCESS GRANTED ✅ **Subtopic (d) Relational:** >= compared Age (20) to the limit. **Subtopic (e) Logical:** Used || (OR) and && (AND) to combine rules.
	Explanation This code checks if a person can get access based on their **age** and **parent permission**. First, the program gets the **age** entered by the user. Then it checks if the **parent permission checkbox** is selected. If the age entered is **not a number**, the program shows a message asking the user to enter a valid age. Next, the program checks the condition: If the **age is 18 or older**, OR the **age is 13 or older and the parent allows it**, then **access is granted**. If these conditions are **not true**, then **access is denied**. Finally, the result (**Access Granted or Access Denied**) is shown on the webpage, and a message explains how the decision was made.

3. Quick Check (Reflection)

Instruction: Answer the following questions based on your coding results.

1. **Modulus Utility:** Based on Task 1, if students % 2 returns 0, is the number even or odd?

   If students % 2 = 0, the number is even.

2. **Shorthand Power:** In Task 2, rewrite total -= 20 using the standard assignment method.

   Standard assignment form of total -= 20 is:
   total = total - 20

3. **Logical Priority:** In Task 4, why did we put (age >= 13 && hasParent) in parentheses?

Parentheses are used to group the conditions together so the program checks age ⩾ 13 AND parent permission first before applying the OR (||) condition.

4. **Time Calculation:** (Extra Challenge) If a variable totalMinutes = 130, write the formula to find the remaining minutes after converting to hours using the **Modulus** operator.

Formula to find the remaining minutes:
remainingMinutes = totalMinutes % 60